

dated, with all inter-core communication happening within the chip.

By contrast, in Intel dual-core processors, the communication has to happen via the front-side bus and the Northbridge on the motherboard, meaning external access.

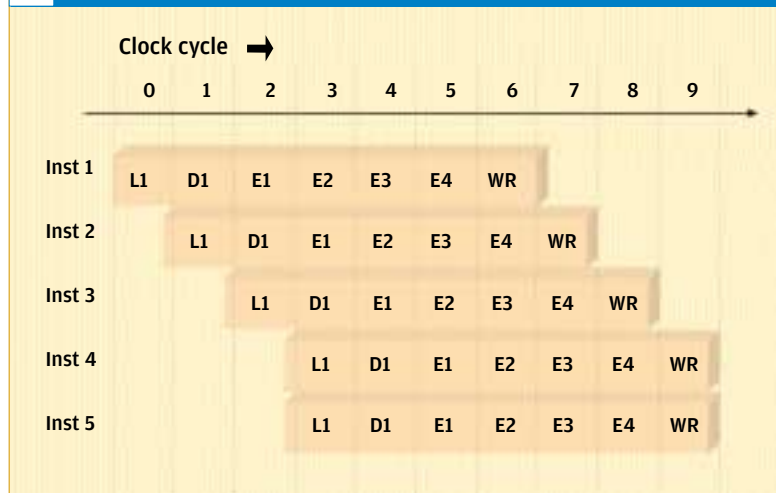
So are AMD's dual-core chips "better and faster" than the Intel one? It seems so. To quote News.com, "Both Anandtech and Tom's Hardware—two hardware benchmarking sites—have published reports stating that, in their own tests, the dual-core Athlon 64 X2 chips generally edge out the Intel Pentium D and the dual-core Pentium 4 Extreme Edition, although the results vary by the tests. Anandtech found that the fastest dual-core Athlon, the 4800+, and often other dual-core Athlons, typically outperformed the Intel chips on tests for single applications, such as running Adobe Photoshop or DivX."

This is not surprising, since it seems that Intel's Pentium D, a dual-core processor, is just like two identical P4 Prescott processor dies tacked together.

## Heat

Heat is the enemy of processors, to the extent that one of the reasons clock speeds can't just be bumped up is heat! How much heat do dual-core processors produce? Fortunately, not too much. AMD says their dual-cores will produce about the same amount of heat as their single-cores. Intel's dual-core processors, like the Pentium D, pro-

## How Pipelining Works



Pipelining is one of the fundamental techniques to speed up processors. The orange blocks above are instructions, broken into their component parts. 'L1' stands for 'Load'; 'D1' stands for 'Decipher'; 'E' stands for 'Execute'. So here, in clock cycle 1, the processor loads instruction 1. In cycle 2, the first instruction is being deciphered, while at the same time, the second instruction is being loaded. Clock cycle 3 sees instruction 1 being executed, instruction 2 being deciphered, and so on. In dual-core processors, each core has its own pipeline.

duce more heat than AMDs—but even this is not alarmingly high.

We've all heard about heating issues with processors, and now, when there are two cores on a single die, shouldn't there be double the amount of heat?

There are a couple of reasons that this is not the case, apart from the obvious fact that not both cores will be maxed out at the same time. Firstly, clock speeds: dual-core processors are, in general, clocked somewhat slower than their single-core brethren. And secondly, dual-core processors are being manufactured using the 90 nm fabrication process, just like the latest single-cores. The "90 nm" refers to the smallest 'feature' size on the die, usually the gate width of the transistors. A smaller fabrication size leads to less power dissipation, and therefore less heat.

## The Northbridge

The Northbridge is one of the two chips in the core logic chipset on a motherboard, the other being the Southbridge. Sometimes—but not too often—the two have been combined on to one die, when design complexity and fabrication processes permitted it. But in general, core logic chipsets are divided into these two main parts. The Northbridge typically handles communications between the CPU, RAM, AGP port or PCI Express, and the Southbridge. Some Northbridges also have integrated video controllers.

In our context, the point is that AMD's dual-cores have a Northbridge-like provision within the chip itself, so there's no need to go to the motherboard for inter-processor communication.

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